SANSKRITI SINGH

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Education

Madan Mohan Malaviya University of Technology — **Sep. 2024 – Present**

B.Tech in Computer Science and Engineering | **CGPA: 8.43** — *Gorakhpur, Uttar Pradesh*

Projects

Tripzy – Real-Time Group Trip Planner | *React, Node.js, Socket.IO, PostgreSQL, Gemini API* — **GitHub** | **Live Link**

- Built a **full-stack** collaborative trip planner with **React** and an **Express** API of **50 REST endpoints** across 9 modules, using **PostgreSQL** and **Prisma**.
- Implemented **real-time collaboration** with **Socket.IO** rooms, broadcasting change events after each committed write so clients re-fetch only updated data, plus **live presence**.
- Developed an **AI itinerary planner** using **Gemini structured JSON** with server-side validation and repair, **model fallbacks** and an automatic route-based backup plan.
- Secured the API with **JWT refresh-token rotation**, **role-based access control**, **rate limiting**, idempotent expense creation and **optimistic concurrency** that rejects stale edits.
- Wrote **179 automated tests** with **Vitest** and **Supertest** against a real database, and deployed on **Vercel**, **Render** and **Neon** with **Docker Compose** locally.

LLMux – Self-Hosted LLM Gateway | *Node.js, Express, Redis, PostgreSQL, React* — **GitHub** | **Live Demo**

- Built a self-hosted **LLM gateway** in Node.js and Express that unifies **OpenAI, Anthropic, Gemini and Ollama** behind one API using the **adapter pattern**.
- Implemented **SSE and WebSocket token streaming** with normalized events, client cancellation via AbortController, **backpressure handling** and fallback before the first token.
- Designed a reliability layer with exponential-backoff retries, fallback chains, **Redis-backed circuit breakers**, response caching and atomic **Lua token-bucket rate limiting**.
- Developed a React dashboard and PostgreSQL usage logging that track tokens, latency and **micro-dollar cost**, tested with Vitest mock providers in **GitHub Actions CI**.

Technical Skills

Programming Languages: C/C++, Java, JavaScript, SQL, HTML/CSS

Technologies/Frameworks: React, Node.js, Express, Socket.IO, Prisma, Zustand, Vitest

Databases: PostgreSQL, Redis

Developer Tools: Git, GitHub, Docker, VS Code, Vercel, Render

Coursework: Data Structures and Algorithms, DBMS, OOPS, Computer Networks, Operating Systems

Achievements

- Solved **400+** problems on LeetCode with a contest rating of **1640** (top **19.6%** globally). Profile Link
- Solved **548** problems on CodeChef as a **2-star** coder with a max rating of **1522**. Profile Link
- Proficient in **Data Structures and Algorithms** and problem solving, built through consistent practice across **900+** problems.